Open Source for Regulated Manufacturing

GMP-Compliant IIoT Data Logging with Apache IoTDB

Lukas Ott otluk@apache.org

Why This Matters

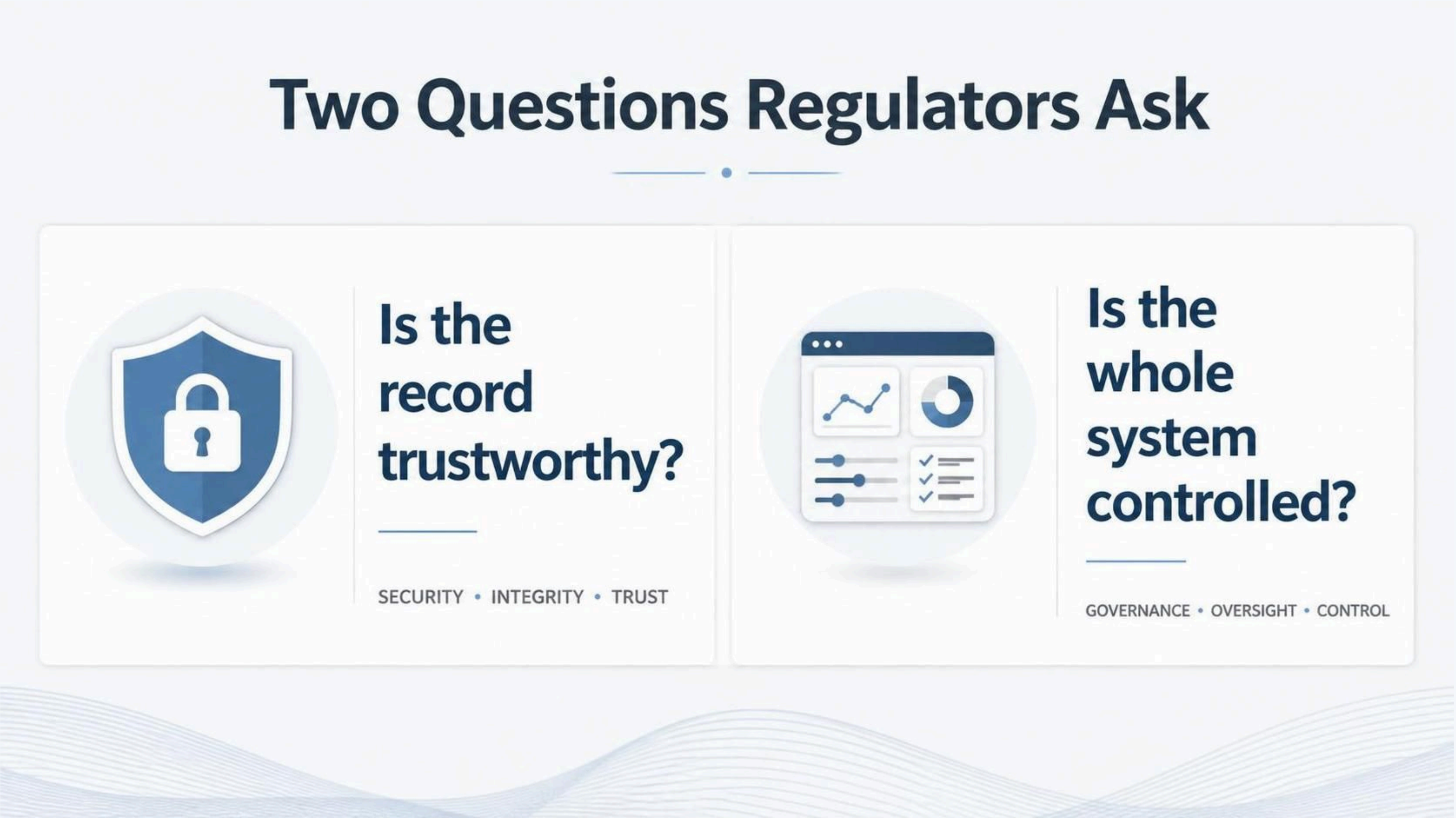
The Manufacturing Reality

- Pharma, MedTech, and Biotech run on high-frequency sensor and machine data.
- That data must be trustworthy, traceable, and available for inspection.
- The challenge is to keep up with velocity without losing compliance.
- The question: can open source become a compliant historian backbone?

The Regulatory Pressure Is Rising

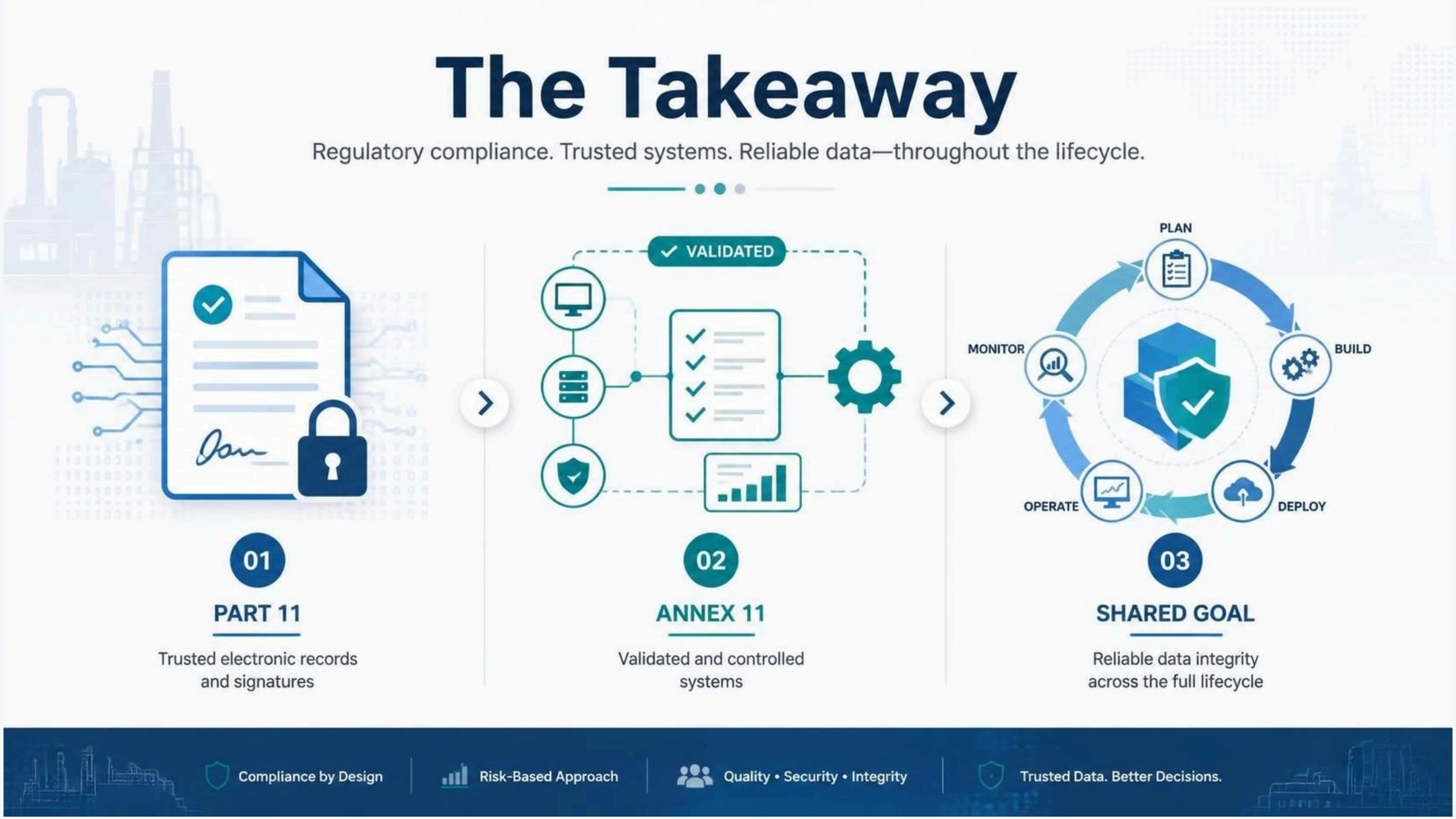
- GMP expectations keep expanding around data integrity, cloud use, and security.
- AI-supported systems are becoming part of regulated manufacturing.
- IIoT platforms now need to handle volume, speed, and auditability together.
- The result: compliance has to be designed in, not added later.

Regulatory Compliance



Comaparing the two Aspects of Regulations

Shared Compliance Goal



What Compliance Really Means for Data

ALCOA++ as the Practical Lens

ALCOA: Core Principles

Principle	Definition
Attributable	Clearly linked to the person or device that generated it
Legible	Readable and accessible throughout its lifecycle
Contemporaneous	Captured at the time of the activity
Original	Preserved as the first capture of information
Accurate	Correct, truthful, and free from tampering

ALCOA+: The Extensions

Principle	Definition
Complete	All changes and deletions remain traceable
Consistent	Data remains reproducible and aligned with expectations
Enduring	Stored on durable media for required retention
Available	Accessible to authorized personnel when needed

ALCOA++ for IIoT

Principle	Definition
Integrity	Protected from unauthorized change
Robustness	Survives high-velocity ingestion without data loss
Transparency	Processing logic can be inspected and reviewed
Accountability	Actions can be traced to responsible users or systems
Reliability	Behaves predictably in validated environments

Why Traditional Historians Struggle

- Proprietary pricing often scales with tags, ingestion, or retention.
- Vendor lock-in makes architecture changes expensive.
- Validation becomes harder when systems are opaque.
- High-frequency streams expose performance and audit gaps.

The challenge is not just storing data. The challenge is storing it in a way that survives inspection, scale, and change.

Why Open Source

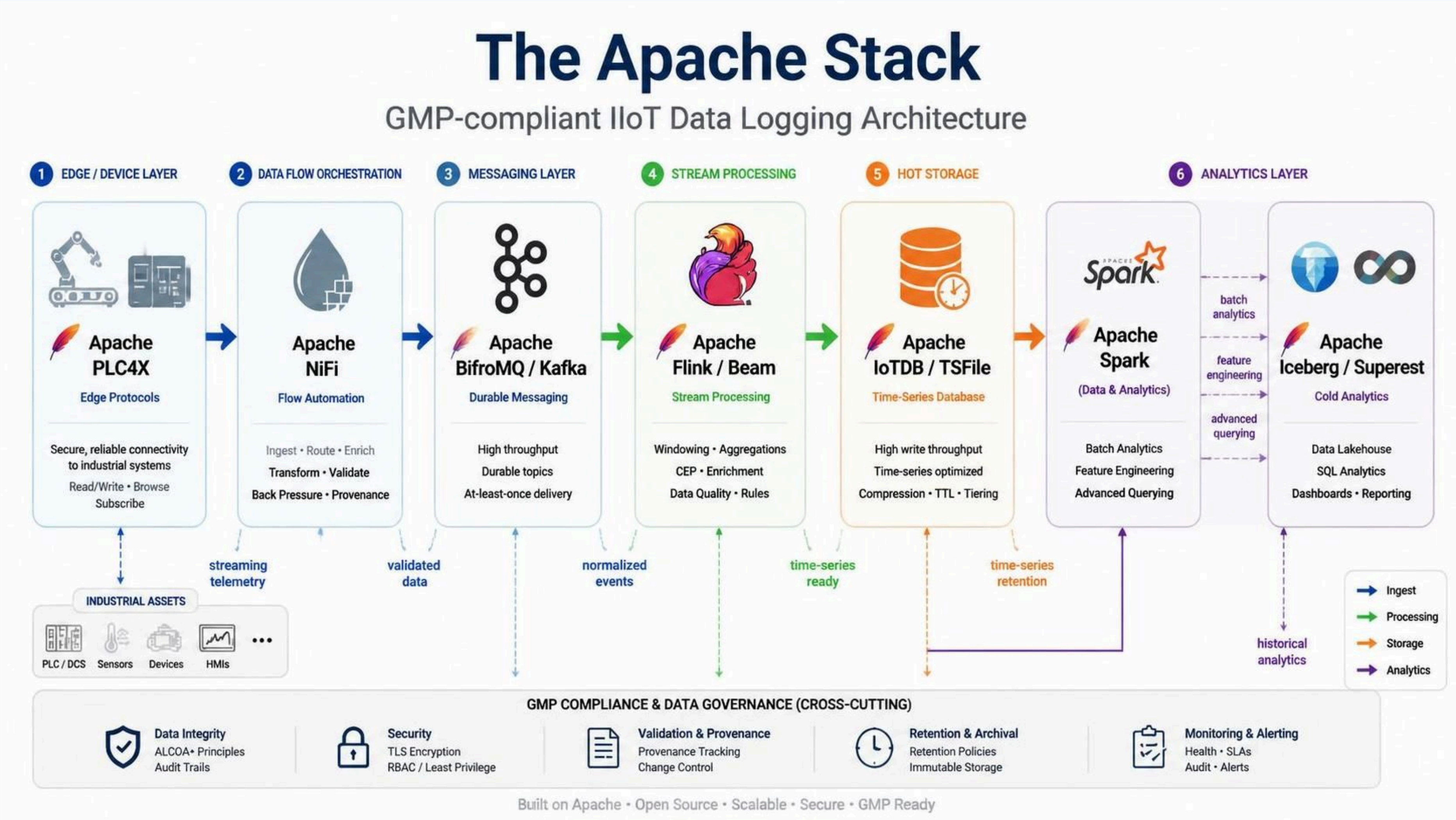
Open Source Fits Regulated Environments

- Transparency: auditors can inspect the logic, not just the marketing.
- Interoperability: open APIs reduce siloed architectures.
- Governance: community-driven projects reduce dependence on a single vendor.
- Flexibility: architecture can evolve with the process, not against it.

A Real-World Reference

- Open infrastructure can support mission-critical preservation and monitoring.
- Apache-based systems have already been recognized in global innovation contexts.
- That matters because regulated industries need proof, not promises.

The Apache Stack



Apache IoTDB as the Historian Core

- Built for high-frequency industrial sensors and PLC time-series.
- Supports auditability, versioning, and lineage-friendly storage.
- Includes authentication, RBAC, encryption, and retention controls.
- Designed for use cases where data volume and compliance both matter.

Apache TSFile as the Storage Foundation

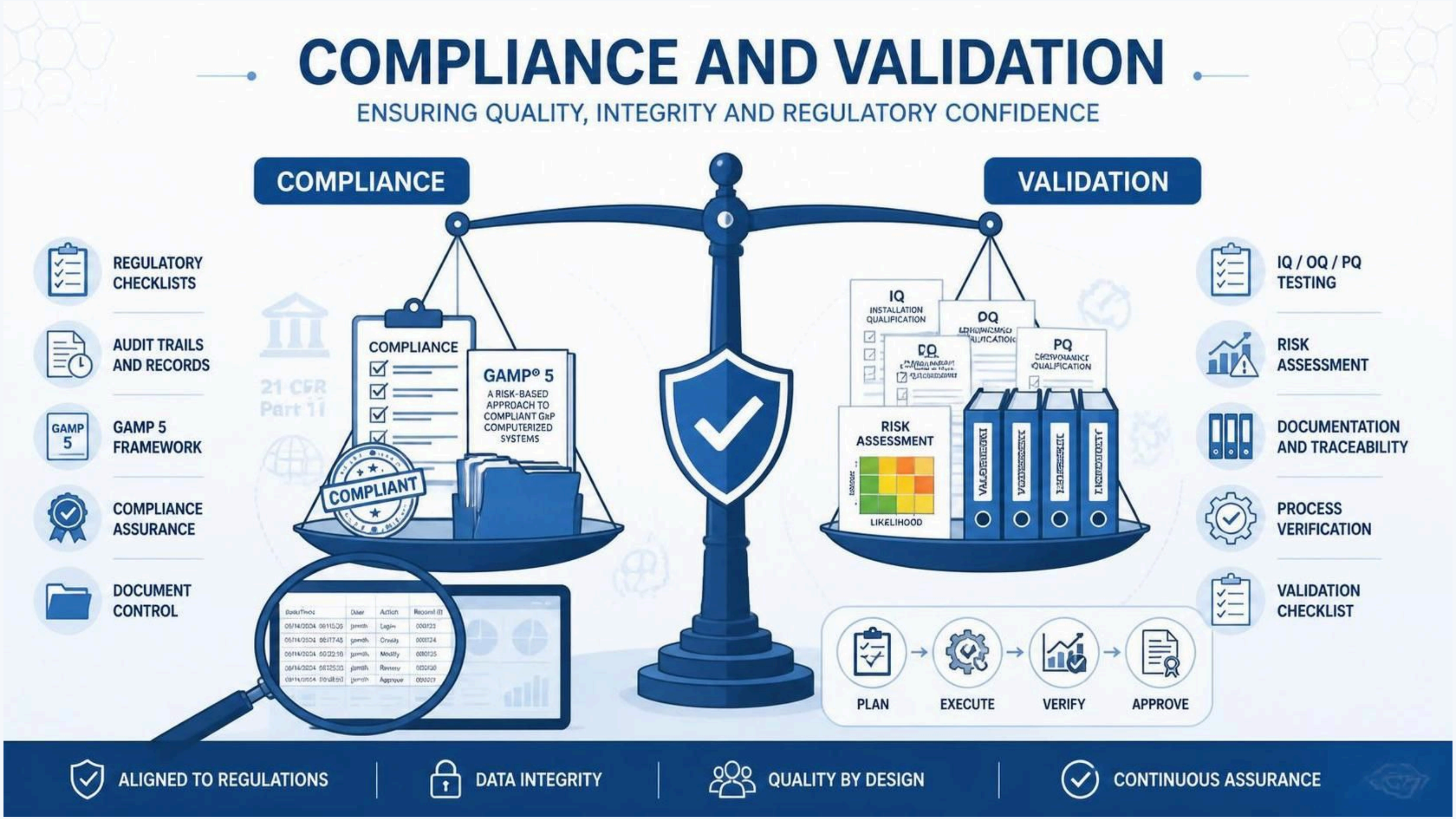
- Append-only by design, which supports immutable storage patterns.
- Columnar layout improves efficiency and validation boundaries.
- Works well with historical reproducibility and spot-checking.
- Can be paired with Apache Iceberg for richer query lineage.

End-to-End Data Flow

- Edge and protocol parsing: Apache PLC 4 X
- Flow orchestration and integration: Apache NiFi
- Messaging and buffering: Apache BifroMQ and Apache Kafka
- Stream processing and rules: Apache Flink or Apache Beam
- Hot storage and history: Apache IoTDB and Apache TSFile
- Cold analytics: Apache Iceberg and Apache Superset

Reference Architecture

Compliance and Validation



Reproducable builds

CSV With GAMP 5

Step	Purpose
1. Risk Assessment	Identify compliance and operational risks early
2. Supplier Qualification	Confirm the platform and vendors are suitable
3. URS	Define user and business requirements
4. Design & Specifications	Translate needs into controlled design
5. IQ/OQ/PQ	Prove installation, operation, and performance
6. Data Integrity Controls	Verify traceability and auditability

What Good Testing Looks Like

Test	What it proves
IQ	The system is deployed repeatably and correctly
OQ	Security, failover, and retention behave as intended
PQ	The system remains stable under peak IIoT load

Code of Federal Regulations Part

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The Demo

Live Demo: Hybrid IoT Data Logging and Visualization

- Goal: show the path from ingestion to buffering to synchronization.
- Watch for authenticated ingestion, offline buffering, and live sync progress.
- The key story: the system keeps running even when the network does not.

Demo Steps

Step	Action
1	Log in as `operator` with `operator`
2	Send sample readings to `/ingest` or use the ingestion helper
3	Open the dashboard and observe the local buffer chart
4	Trigger `Sync to IoTDB` and watch the live SSE progress stream
5	Verify the synchronized historical data appears in the IoTDB chart

Conclusion

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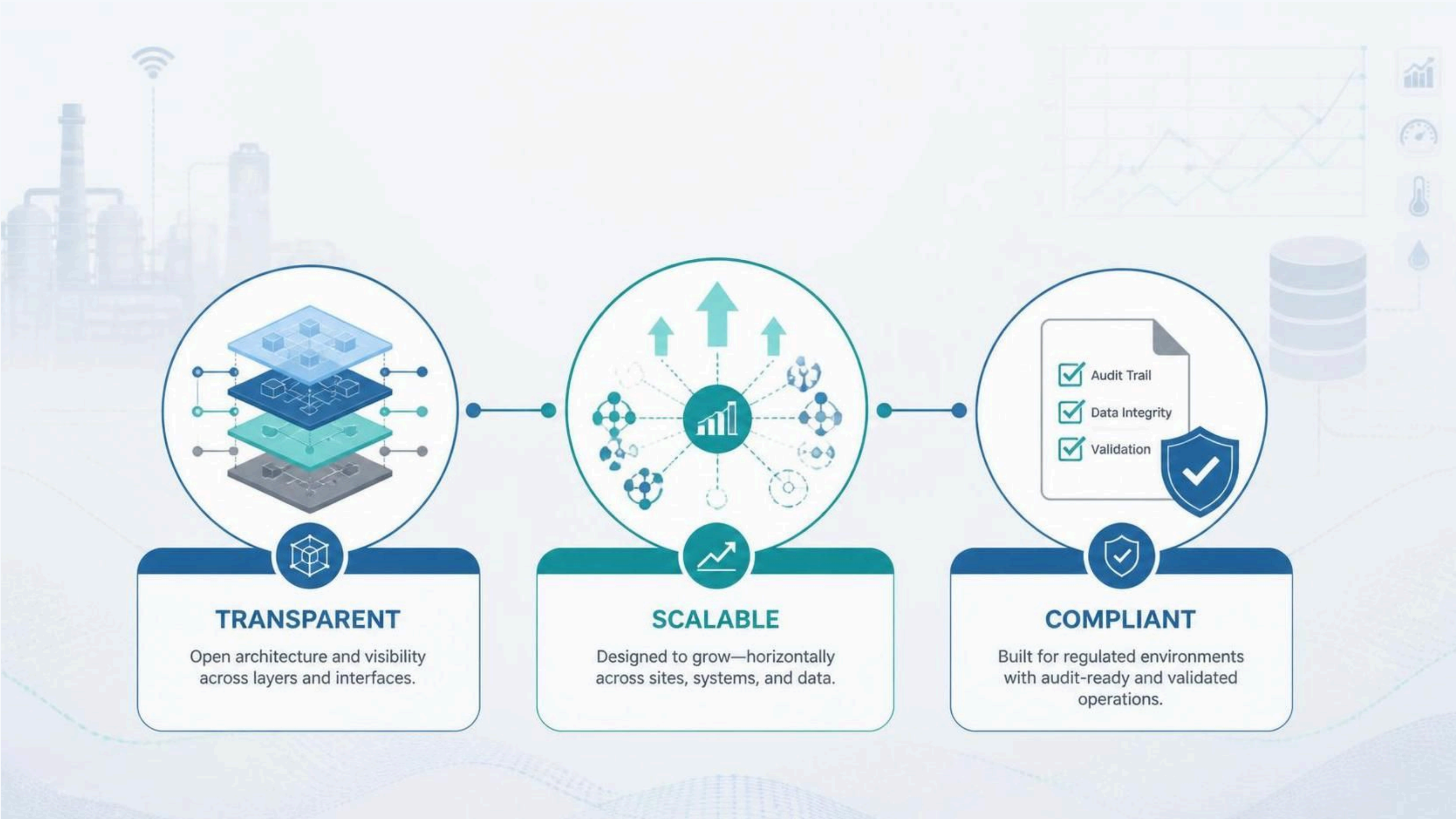
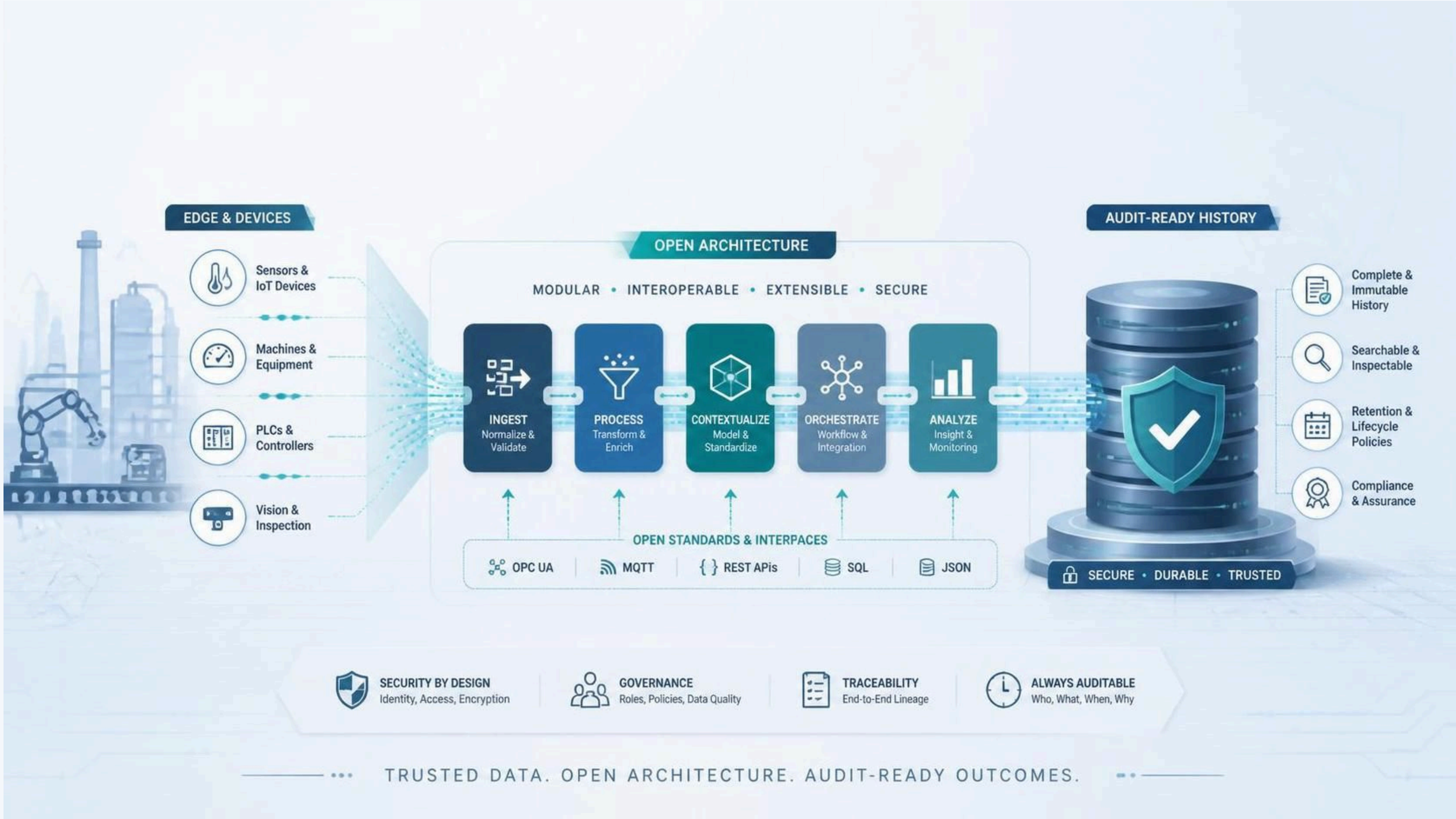
Building a trustworthy, scalable, and compliant data future—together.
EDGE DATA
Real-time. Distributed.
Always on.
SECURE TRACEABILITY BRIDGE
Connecting data today to trusted decisions tomorrow.
AUDIT-READY HISTORY
Verifiable. Immutable.
Always ready.
OPEN SOURCE
Transparent.
Collaborative.
Built for everyone.
COMPLIANCE
Standards-aligned.
Policy-driven.
Risk-aware.
SCALABILITY
Built to grow.
Performance-ready.
Future-proof.
TRUSTWORTHINESS
Verifiable.
Consistent.
Dependable.
DATA INTEGRITY
Accurate. Consistent. Tamper-evident.
Trusted at every step.
SYSTEM RELIABILITY
Resilient. Available. Dependable.
Built for continuous operation.
A solid foundation. A clear path.
A better future—together.

What to Remember



Final Message



Q&A

- Thank you.
- Projects:
  - Apache IoTDB: https://iotdb.apache.org/
  - Apache PLC4X: https://plc4x.apache.org/
  - Apache StreamPipes: https://streampipes.apache.org/
- otluk@apache.org